Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force Output compare: Writing 1 to it forces the wave generator to act as if a compare match has occurred.

WGM01:00 D6 D3 Timer0 mode selector bits

D6	D3	
0	0	Normal
0	1	CTC (Clear Timer on Compare match)
1	0	PWM, phase correct
1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode; The table shows what the wave generator does on compare match when the timer is in Normal or CTC mode:

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

CS02:00 D2D1D0 Timer0 clock selector

D2	D1	D0	
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (no prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge
1	1	1	External clock source on T0 pin. Clock on rising edge

Figure 15-4. TCCR0 (Timer/Counter Control Register) Register

Bit	7	6	5	4	3	2	1	0
	COM1A1	COM1A0	COM1B1	COM1B0	FOC1A	FOC1B	WGM11	WGM10
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

COM1A1:COM1A0 D7 D6 Compare Output Mode for Channel A (discussed in Section 9-3)

COM1B1:COM1B0 D5 D4 Compare Output Mode for Channel B (discussed in Section 9-3)

FOC1A D3 Force Output Compare for Channel A (discussed in Section 9-3)

FOC1B D2 Force Output Compare for Channel B (discussed in Section 9-3)

WGM11:10 D1 D0 Timer1 mode (discussed in Figure 9-18)

Figure 9-17. TCCR1A (Timer 1 Control) Register

Bit	7	6	5	4	3	2	1	0	
	ICNC1	ICES1	-	WGM13	WGM12	CS12	CS11	CS10	TCCR1B
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ICNC1 D7 Input Capture Noise Canceler
0 = Input Capture is disabled.
1 = Input Capture is enabled.

ICES1 D6 Input Capture Edge Select
0 = Capture on the falling (negative) edge
1 = Capture on the rising (positive) edge

D5 Not used

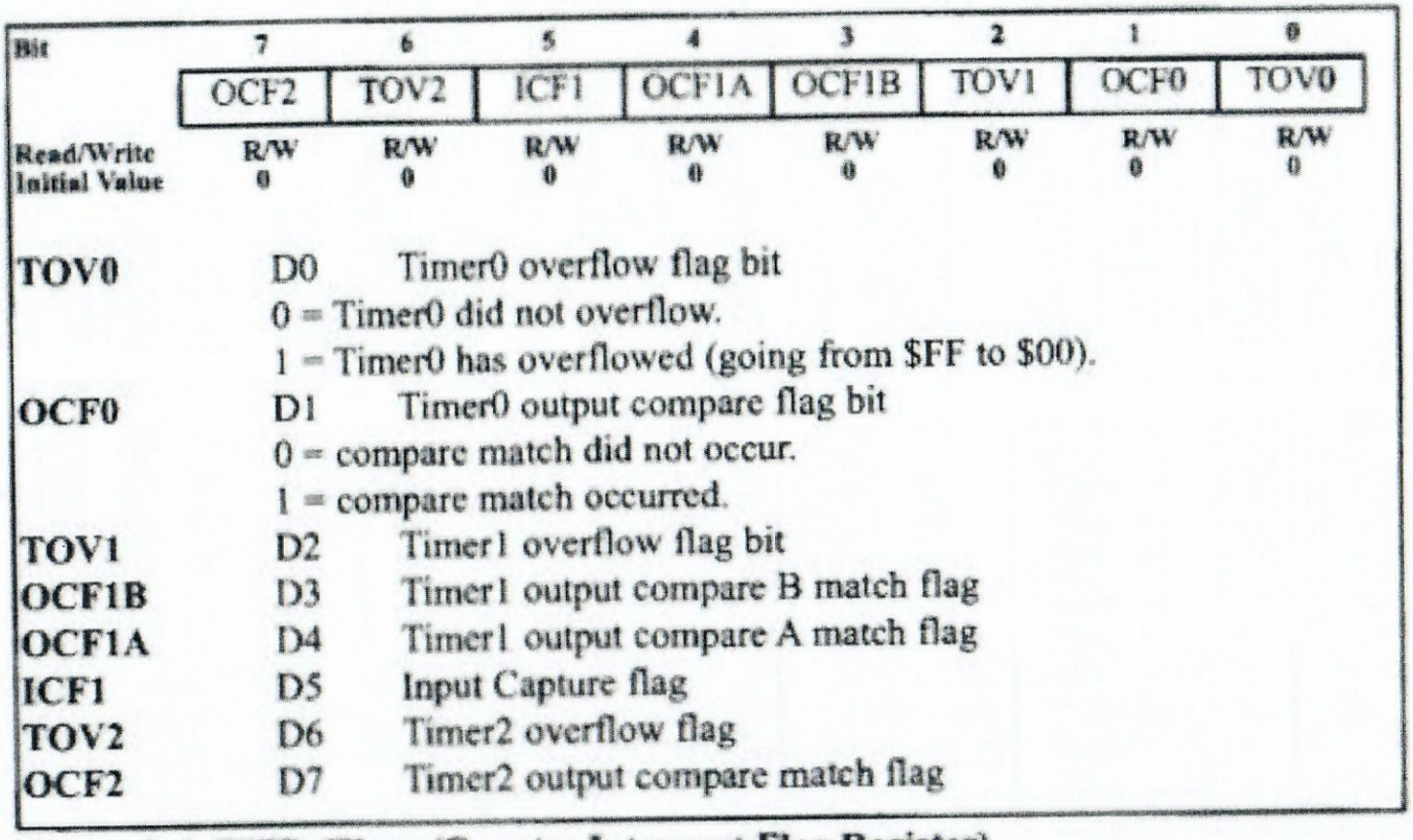
WGM13:WGM12 D4 D3 Timer1 mode

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	-	-	-
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

CS12:CS10 D2D1D0 Timer1 clock selector

0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (no prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T1 pin. Clock on falling edge.
1	1	1	External clock source on T1 pin. Clock on rising edge.

Figure 9-18. TCCR1B (Timer 1 Control) Register



Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

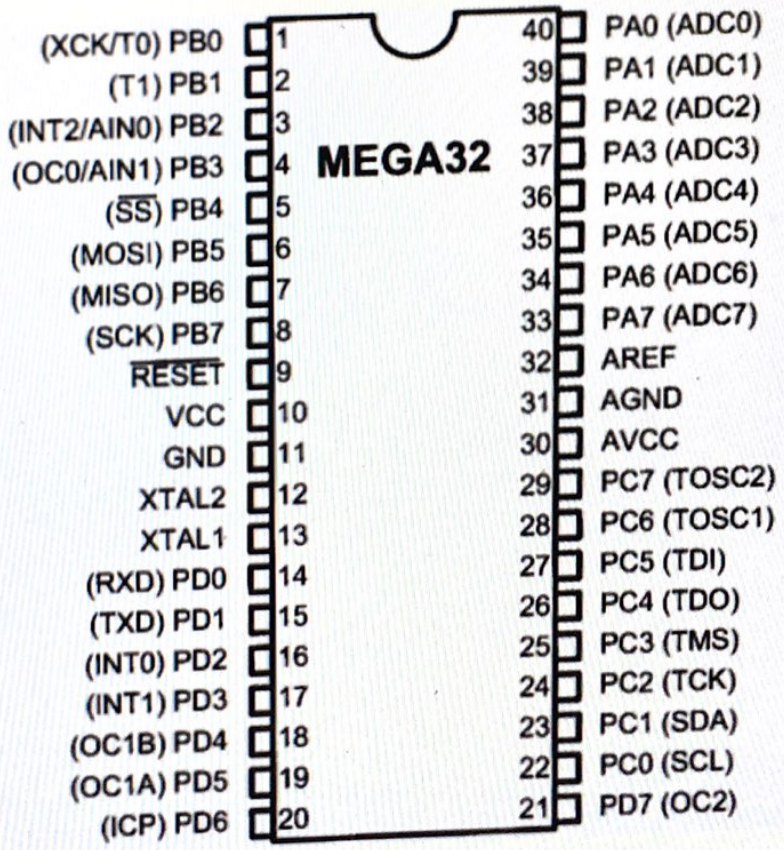
ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)

Chapter 4-Fig4-1



Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force compare match: This is a write-only bit, which can be used while generating a wave. Writing 1 to it causes the wave generator to act as if a compare match had occurred.

WGM00, WGM01

D6	D3	Timer0 mode selector bits
0	0	Normal
0	1	CTC (Clear Timer on Compare Match)
1	0	PWM, phase correct
1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode:
These bits control the waveform generator (see Chapter 15).

CS02:00

D2	D1	D0	Timer0 clock selector
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (No Prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

Figure 9-5. TCCR0 (Timer/Counter Control Register) Register

Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)

D7							D0
OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0

TOIE0 Timer0 overflow interrupt enable
= 0 Disables Timer0 overflow interrupt
= 1 Enables Timer0 overflow interrupt

OCIE0 Timer0 output compare match interrupt enable
= 0 Disables Timer0 compare match interrupt
= 1 Enables Timer0 compare match interrupt

TOIE1 Timer1 overflow interrupt enable
= 0 Disables Timer1 overflow interrupt
= 1 Enables Timer1 overflow interrupt

OCIE1B Timer1 output compare B match interrupt enable
= 0 Disables Timer1 compare B match interrupt
= 1 Enables Timer1 compare B match interrupt

OCIE1A Timer1 output compare A match interrupt enable
= 0 Disables Timer1 compare A match interrupt
= 1 Enables Timer1 compare A match interrupt

TICIE1 Timer1 input capture interrupt enable
= 0 Disables Timer1 input capture interrupt
= 1 Enables Timer1 input capture interrupt

TOIE2 Timer2 overflow interrupt enable
= 0 Disables Timer2 overflow interrupt
= 1 Enables Timer2 overflow interrupt

OCIE2 Timer2 output compare match interrupt enable
= 0 Disables Timer2 compare match interrupt
= 1 Enables Timer2 compare match interrupt

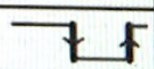
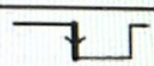
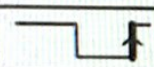
These bits, along with the I bit, must be set high for an interrupt to be responded to. Upon activation of the interrupt, the I bit is cleared by the AVR itself to make sure another interrupt cannot interrupt the microcontroller while it is servicing the current one. At the end of the ISR, the RETI instruction will make I = 1 to allow another interrupt to come in.

Figure 10-3. TIMSK (Timer Interrupt Mask) Register

D7 D0

SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00

ISC01, ISC00 (Interrupt Sense Control bits) These bits define the level or edge on the external INT0 pin that activates the interrupt, as shown in the following table:

ISC01	ISC00		Description
0	0		The low level of INT0 generates an interrupt request.
0	1		Any logical change on INT0 generates an interrupt request.
1	0		The falling edge of INT0 generates an interrupt request.
1	1		The rising edge of INT0 generates an interrupt request.

ISC11, ISC10 These bits define the level or edge that activates the INT1 pin.

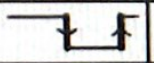
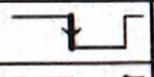
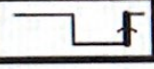
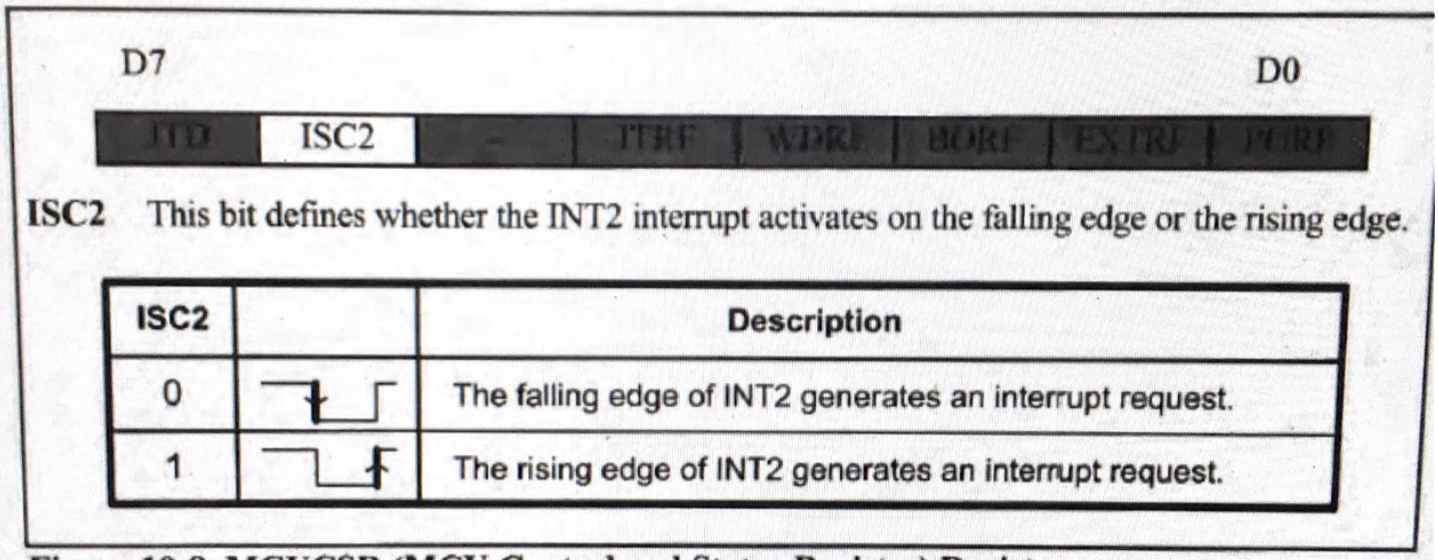
ISC11	ISC10		Description
0	0		The low level of INT1 generates an interrupt request.
0	1		Any logical change on INT1 generates an interrupt request.
1	0		The falling edge of INT1 generates an interrupt request.
1	1		The rising edge of INT1 generates an interrupt request.

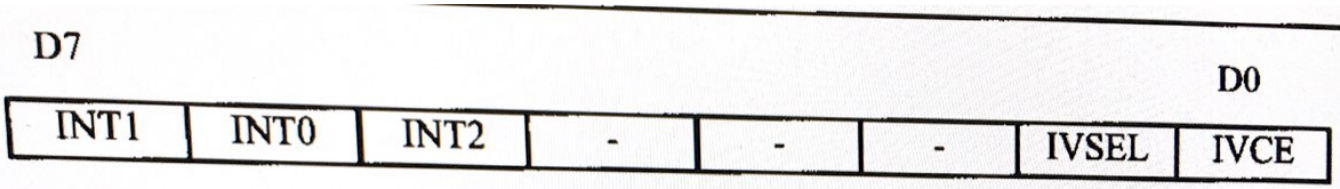
Figure 10-7. MCUCR (MCU Control Register) Register



ISC2 This bit defines whether the INT2 interrupt activates on the falling edge or the rising edge.

ISC2		Description
0		The falling edge of INT2 generates an interrupt request.
1		The rising edge of INT2 generates an interrupt request.

Figure 10-8. MCUCSR (MCU Control and Status Register) Register



INT0 External Interrupt Request 0 Enable
= 0 Disables external interrupt 0
= 1 Enables external interrupt 0

INT1 External Interrupt Request 1 Enable
= 0 Disables external interrupt 1
= 1 Enables external interrupt 1

INT2 External Interrupt Request 2 Enable
= 0 Disables external interrupt 2
= 1 Enables external interrupt 2

These bits, along with the I bit, must be set high for an interrupt to be responded to.

Figure 10-6. GICR (General Interrupt Control Register) Register

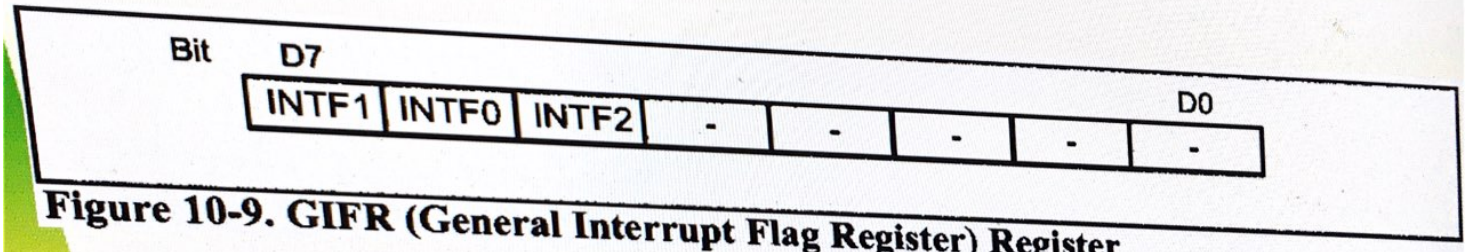


Figure 10-9. GIFR (General Interrupt Flag Register) Register

Atmega32 Interrupt Vector Table

Table 10-1: Interrupt Vector Table for the ATmega32 AVR

Interrupt	ROM Location (Hex)
Reset	0000
External Interrupt request 0	0002
External Interrupt request 1	0004
External Interrupt request 2	0006
Time/Counter2 Compare Match	0008
Time/Counter2 Overflow	000A
Time/Counter1 Capture Event	000C
Time/Counter1 Compare Match A	000E
Time/Counter1 Compare Match B	0010
Time/Counter1 Overflow	0012
Time/Counter0 Compare Match	0014
Time/Counter0 Overflow	0016
SPI Transfer complete	0018
USART, Receive complete	001A
USART, Data Register Empty	001C
USART, Transmit Complete	001E
ADC Conversion complete	0020
EEPROM ready	0022
Analog Comparator	0024
Two-wire Serial Interface (I2C)	0026
Store Program Memory Ready	0028

ATmega324pb Interrupt Vector Table

14.1 Interrupt Vectors in ATmega324PB

Table 14-1. Reset and Interrupt Vectors in ATmega324PB

Vector No	Program Address(2)	Source	Interrupts definition
1	0x0000(1)	RESET	External Pin, Power-on Reset, Brown-out Reset and Watchdog System Reset
2	0x0002	INT0	External Interrupt Request 0
3	0x0004	INT1	External Interrupt Request 1
4	0x0006	INT2	External Interrupt Request 2
5	0x0008	PCINT0	Pin Change Interrupt Request 0
6	0x000A	PCINT1	Pin Change Interrupt Request 1
7	0x000C	PCINT2	Pin Change Interrupt Request 2
8	0x000E	PCINT3	Pin Change Interrupt Request 3
9	0x0010	WDT	Watchdog Time-out Interrupt
10	0x0012	TIMER2_COMPA	Timer/Counter2 Compare Match A
11	0x0014	TIMER2_COMPB	Timer/Coutner2 Compare Match B
12	0x0016	TIMER2_OVF	Timer/Counter2 Overflow
13	0x0018	TIMER1_CAPT	Timer/Counter1 Capture Event
14	0x001A	TIMER1_COMPA	Timer/Counter1 Compare Match A
15	0x001C	TIMER1_COMPB	Timer/Coutner1 Compare Match B
16	0x001E	TIMER1_OVF	Timer/Counter1 Overflow
17	0x0020	TIMER0_COMPA	Timer/Counter0 Compare Match A
18	0x0022	TIMER0_COMPB	Timer/Coutner0 Compare Match B
19	0x0024	TIMER0_OVF	Timer/Counter0 Overflow
20	0x0026	SPI0_STC	SPI0 Serial Transfer Complete
21	0x0028	USART0_RX	USART0 Rx Complete
22	0x002A	USART0_UDRE	USART0, Data Register Empty

http://ww1.microchip.com/downloads/en/DeviceDoc/40001908A.pdf

Table 10-3: Interrupt Vector Name for the ATmega32/ATmega16 in WinAVR

Interrupt	Vector Name in WinAVR
External Interrupt request 0	INT0_vect
External Interrupt request 1	INT1_vect
External Interrupt request 2	INT2_vect
Time/Counter2 Compare Match	TIMER2_COMP_vect
Time/Counter2 Overflow	TIMER2_OVF_vect
Time/Counter1 Capture Event	TIMER1_CAPT_vect
Time/Counter1 Compare Match A	TIMER1_COMPA_vect
Time/Counter1 Compare Match B	TIMER1_COMPB_vect
Time/Counter1 Overflow	TIMER1_OVF_vect
Time/Counter0 Compare Match	TIMER0_COMP_vect
Time/Counter0 Overflow	TIMER0_OVF_vect
SPI Transfer complete	SPI_STC_vect
USART, Receive complete	USART0_RX_vect
USART, Data Register Empty	USART0_UDRE_vect
USART, Transmit Complete	USART0_TX_vect
ADC Conversion complete	ADC_vect
EEPROM ready	EE_RDY_vect
Analog Comparator	ANALOG_COMP_vect
Two-wire Serial Interface	TWI_vect
Store Program Memory Ready	SPM_RDY_vect

Address		Name
Mem.	I/O	
$20	$00	TWBR
$21	$01	TWSR
$22	$02	TWAR
$23	$03	TWDR
$24	$04	ADCL
$25	$05	ADCH
$26	$06	ADCSRA
$27	$07	ADMUX
$28	$08	ACSR
$29	$09	UBRRL
$2A	$0A	UCSRB
$2B	$0B	UCSRA
$2C	$0C	UDR
$2D	$0D	SPCR
$2E	$0E	SPSR
$2F	$0F	SPDR
$30	$10	PIND
$31	$11	DDRD
$32	$12	PORTD
$33	$13	PINC
$34	$14	DDRC
$35	$15	PORTC

Address		Name
Mem.	I/O	
$36	$16	PINB
$37	$17	DDRB
$38	$18	PORTB
$39	$19	PINA
$3A	$1A	DDRA
$3B	$1B	PORTA
$3C	$1C	EECR
$3D	$1D	EEDR
$3E	$1E	EEARL
$3F	$1F	EEARH
$40	$20	UBRRC
		UBRRH
$41	$21	WDTCR
$42	$22	ASSR
$43	$23	OCR2
$44	$24	TCNT2
$45	$25	TCCR2
$46	$26	ICR1L
$47	$27	ICR1H
$48	$28	OCR1BL
$49	$29	OCR1BH
$4A	$2A	OCR1AL

Address		Name
Mem.	I/O	
$4B	$2B	OCR1AH
$4C	$2C	TCNT1L
$4D	$2D	TCNT1H
$4E	$2E	TCCR1B
$4F	$2F	TCCR1A
$50	$30	SFIOR
$51	$31	OCDR
		OSCCAL
$52	$32	TCNT0
$53	$33	TCCR0
$54	$34	MCUCSR
$55	$35	MCUCR
$56	$36	TWCR
$57	$37	SPMCR
$58	$38	TIFR
$59	$39	TIMSK
$5A	$3A	GIFR
$5B	$3B	GICR
$5C	$3C	OCR0
$5D	$3D	SPL
$5E	$3E	SPH
$5F	$3F	SREG

Note: Although memory address $20-$5F is set aside for I /O registers (SFR) we can access them as I /O locations with addresses starting at $00.

Figure 2-7. I/O Registers of the ATmega32 and Their Data Memory Address Locations

Ch 10 - Interrupts

polling - microcontroller constantly monitors the status of a given device
- ⇒ when the condition changes/met service is performed
- ⇒ after done service go back to polling process of service next condition

interrupt (event) - a hardware induced subroutine call
- ⇒ devices send an interrupt signal when they need the microcontroller's service
  - ⇒ microcontroller stops what it's doing and services the device through Interrupt service Routine (ISR) or Interrupt Handler
- ⇒ can do other things/tasks while waiting for device requests
- ⇒ every interrupt source must have an ISR or Interrupt Handler

AVR Interrupt Sources

1) Timers (0, 1, 2) - at least 2 interrupts for each
   ① TOVn (timer overflow) 0xFF ⇒ 0x00
   ② Compare match OCFn ⇒ TCNTn = OCRn
2) 3 Interrupts for external hardware
   - INT0 ⇒ PD2
   - INT1 ⇒ PD3
   - INT2 ⇒ PB2
3) USART (Universal synchronous/asynchro Receiver Transmission)

⇒ 4) SPI interrupt / I2C
⇒ 5) ADC

SREG ⇒ I, T

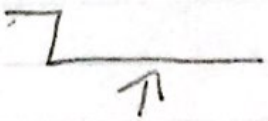
I - global interrupt enable

External Hardware Interrupts

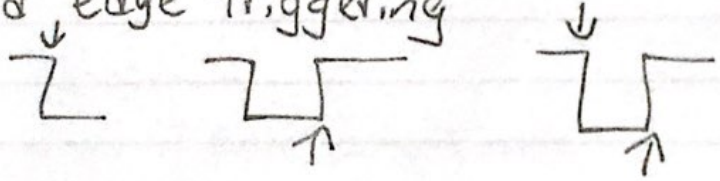
3 hardware interrupts

INT0 (PORTD.2) (PD2)
INT1 (PD3)
INT2 (PB2)

External interrupts can be triggered/ activated by level triggering



and edge triggering



ATmega32 | MCUCR (Microcontroller Unit) Control Register
INT0, INT1

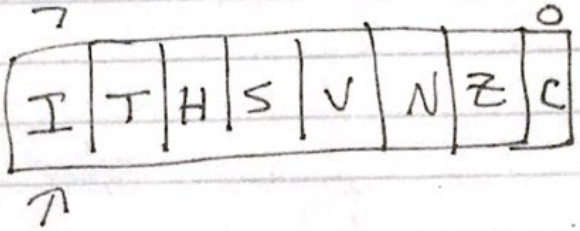
MCUCSR (Microcontroller Unit) Control & Status Register
INT2

Upon reset all interrupt sources are disabled

⇒ microcontroller will not respond to interrupt if activated

steps: (enabling an interrupt source)

1)



I - Global interrupt enable

Assembly: SEI ; I=1 in SREG
C: sei();

disable interrupts
assembly: CLI ; I=0 in SREG
C: cli();

2) if I=1, set (HIGH) interrupt enable (IE) flag for interrupt sources

TIMSK (ATmega32)

Timer Interrupt Flag Bits and Associated Registers

	overflow bit	IE
Timer0	TOV0	TOIE0
Timer1	TOV1	TOIE1
Timer2	TOV2	TOIE2

TIMSK
timer interrupt Mask

Program Ex
PORTC is connected to 8 switches and PORTD to 8 LEDs. Write a program that uses Timer0 to generate a square wave on PB5 while at the same time transferring data from PORTC to PORTD.

PORTC ~~[illegible]~~ input - switches } polling
PORTD - output - LEDs }

Timer0 ⇒ square wave ⇒ interrupt ⇒ OVR
TOV0
OCF0

```
.ORG 0x0; location for reset
JMP MAIN

.ORG 0x16 ; location for Timer0 overflow
            Vector address
JMP TO_OVR_ISR ; jump to ISR for
               ; Timer0

   .ORG 0x100
MAIN: LDI  R20, HIGH(RAMEND)
      OUT  SPH, R20
      LDI  R20, LOW(RAMEND)
      OUT  SPL, R20 ; move SP

      SBI DDRB, 5 ; PB5 as output bit
      LDI R20, (1<<TOIE0)
      OUT TIMSK, R20 ; enable Timer0
                     ; overflow interrupt
      SEI            ; I=1 - Global Interrupt
                     Enable

      LDI  R20, -32; time value for 4ms
                     fosc= 8MHz

      OUT TCNT0, R20; preload for Timer0
      LDI R20, 0x01; 00000001, Timer0,
                     clk, Normal mode, no
      OUT TCCR0, R20; prescale
```

0-63		
~~0-31~~		
I/O	IN	
	OUT	

```
        LDI R20, 0x00 ;
        OUT DDRC, R20 ; PortC input
before  LDI R20, 0xFF
timer0  OUT PORTC, R20 ; pull up resistors PORTC
config  OUT DDRD, R20 ; PORTD output

;----main loop
HERE:   IN  R20, PINC
        OUT PORTD, R20 ; PORTD = PORTC
        RJMP HERE      ; continuously

TOVF    .ORG 0x200                    PB5
TO_OVR_ISR: IN  R16, PORTB            ↓
        LDI R17, 0x20 ; 0010 0000
        EOR R16, R17 ; toggle PB5
        OUT PORTB, R16 ; output PB5
        LDI R16, -32 ;
        OUT TCNT0, R16 ; reload TCNT0
                       ; preload for
                       ; 4 us for
                       ; half wave
        RETI           ; return from ISR
```

do not have to turn off timer 0 to reset TOVF -done by RETI

RET

I=0